DS-A3: Regression Baseline and Diagnostics

1. Dataset and experimental design

The scikit-learn Diabetes dataset has 442 rows, ten numeric baseline inputs, and no missing feature or target values. The target is a quantitative measure of progression one year later. Raw inputs are loaded with scaled=False. Python 3.12.14; scikit-learn 1.7.2.

A frozen 80/20 random split (seed 42) gives 353 training and 89 test rows. Five shuffled training folds (seed 42) compare Dummy, Linear, Ridge and Lasso by mean MAE. The mean Dummy uses only training targets. StandardScaler stays inside each linear pipeline. Ridge and Lasso search alpha 0.1, 1, 10, 100. All ten original features are kept. The choice is frozen before test evaluation.

```
data = load_diabetes(scaled=False, as_frame=True)
Pipeline([('scale', StandardScaler()), ('reg', Ridge())])
GridSearchCV(..., scoring='neg_mean_absolute_error', cv=cv_splits)
```

2. Model comparison

Model	alpha	CV_MAE	CV_SD	Test_MAE	Test_RMSE	Test_R2
Dummy	-	66.867	5.632	64.006	73.222	-0.012
Linear	-	45.048	1.773	42.794	53.853	0.453
Ridge	1.000	44.999	1.808	42.812	53.777	0.454
Lasso	0.100	44.956	1.837	42.805	53.709	0.456

CV_SD is the fold standard deviation; it is not a confidence interval. MAE and RMSE are in target units; R-squared is unitless. Alpha is not applicable for Dummy and Linear. All eight alpha configurations and fold scores are retained in main.ipynb.

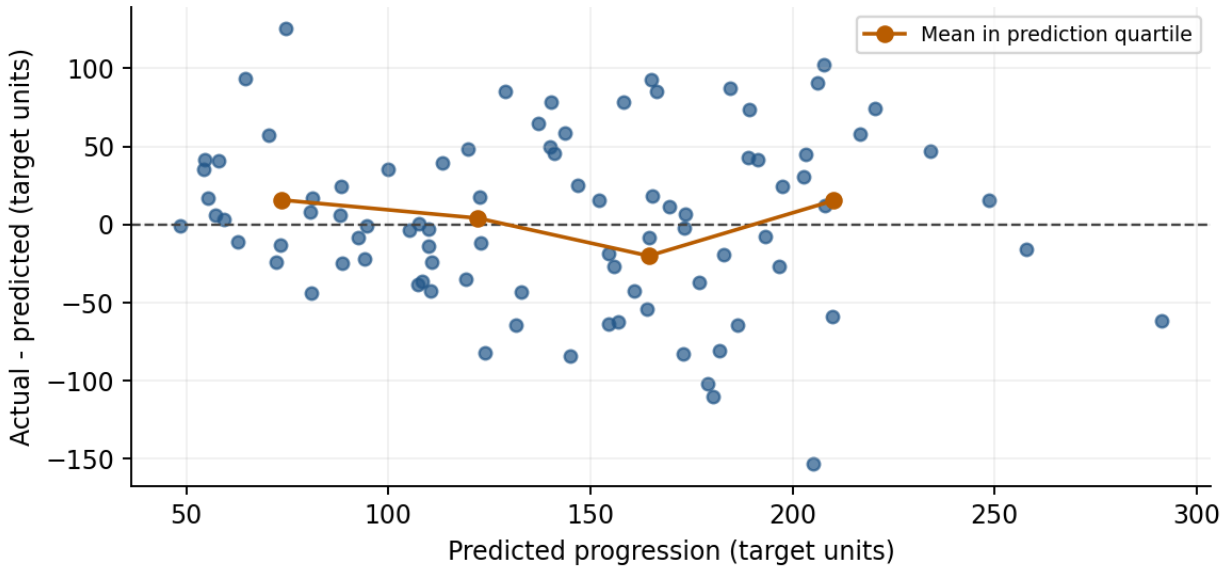
Lasso was selected using CV MAE 44.956. Its test MAE is 42.805, RMSE is 53.709, and R-squared is 0.456. The test ranking is not used for model selection.

Ridge improves CV MAE over ordinary linear regression by 0.049 (0.11%), and Lasso by 0.093 (0.21%). A positive value means lower error. The gains are too small here to claim a meaningful practical improvement. These differences must be judged against the fold variation in the table; this experiment does not establish a statistically reliable improvement. Both add a tuning choice; Lasso can also set coefficients to zero, but the selected Lasso retains 10 of ten coefficients.

Full code, outputs, audit hashes and tuning results: main.ipynb. Per-row test predictions for all four models: results.csv.

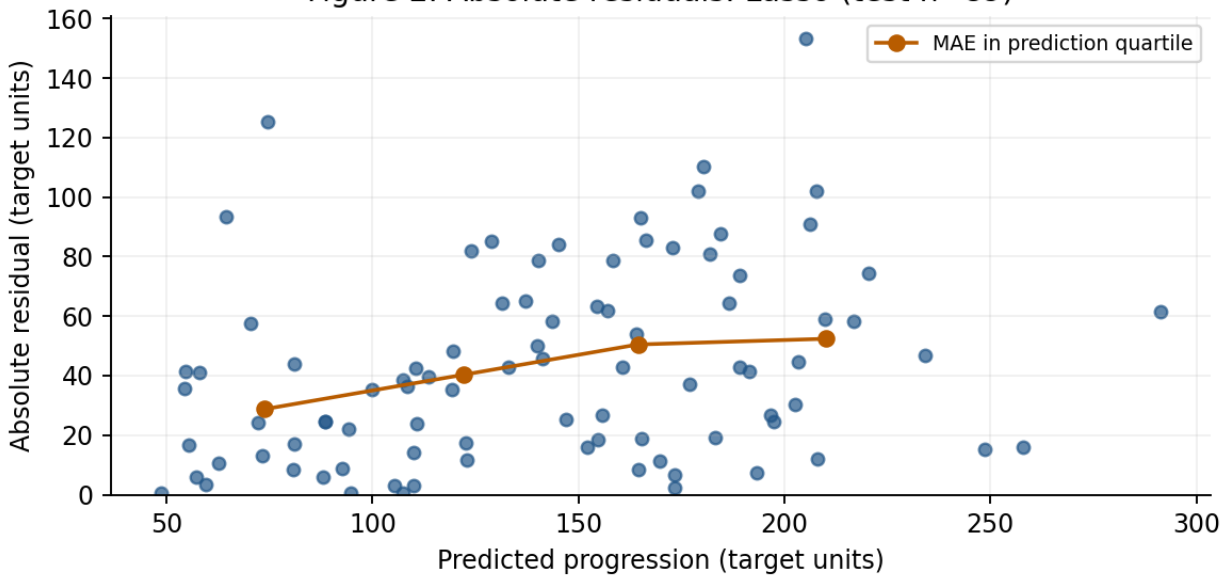
3. Test residual diagnostics

Figure 1. Residuals vs predicted: Lasso (test n=89)



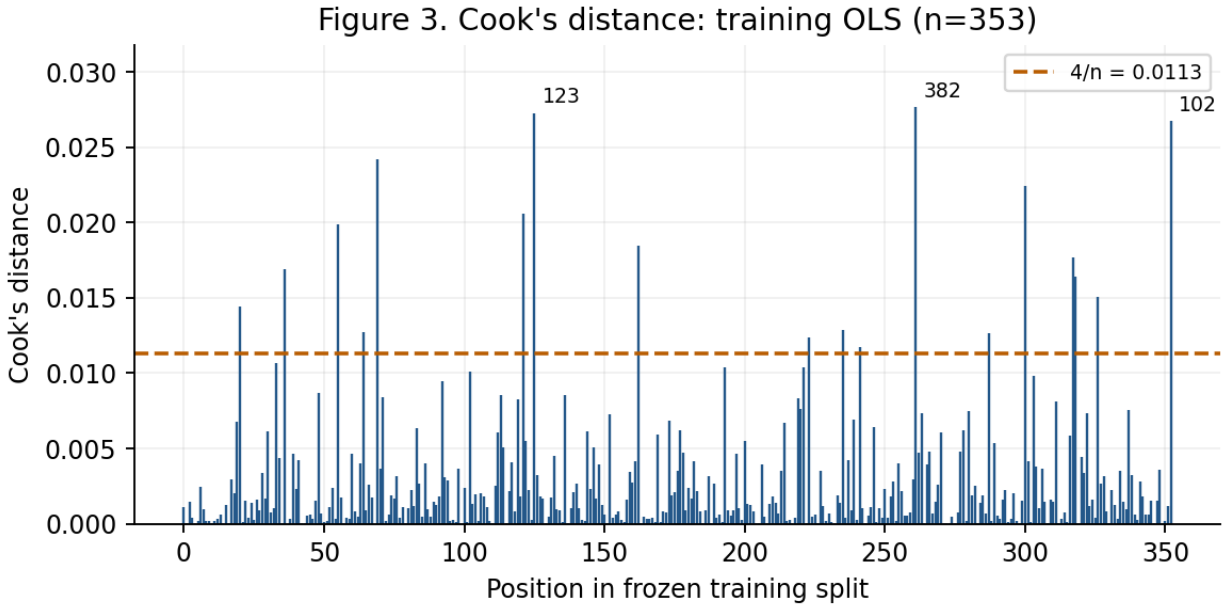
The mean test residual is 3.96. The lowest and highest prediction bins have mean residuals 15.77 and 15.39. The third bin has mean residual -20.02, indicating local overprediction. The low and high bins show underprediction on average. There is substantial scatter around zero. Bin means help check local bias, but 89 test rows cannot establish a stable calibration pattern.

Figure 2. Absolute residuals: Lasso (test n=89)

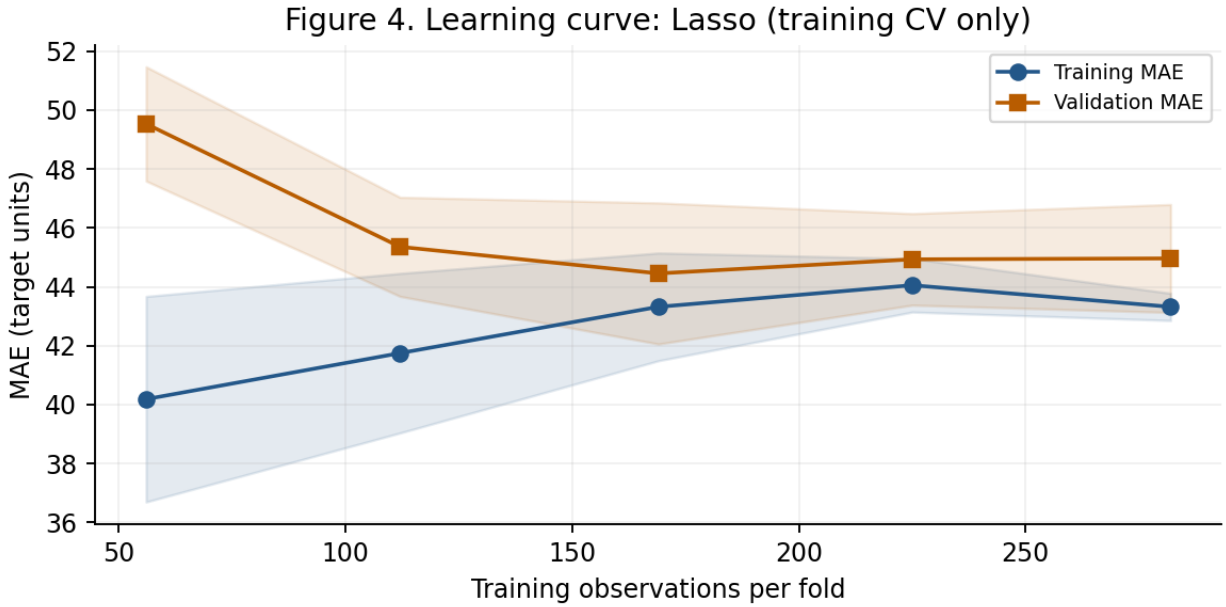


MAE across prediction quartiles ranges from 28.71 to 52.38. The size of errors varies across the prediction range, so constant variance should not be assumed. The pattern is descriptive evidence, not proof of heteroscedasticity or a reason to retune on this test set.

4. Training influence and learning curve



18 training rows exceed 4/n = 0.0113. The largest distance is 0.0277, at original row 382. Influential observations can move the OLS coefficients and fitted values. They are not automatically incorrect. No rows are removed; this OLS measure is not an exact influence measure for a penalized final model.



Validation MAE changes from 49.53 at 56 rows to 44.96 at 282 rows per fold. At the largest size, training MAE is 43.32, leaving a 1.64-unit gap. The small final gap does not suggest severe overfitting. Validation error levels off after the first few sizes. More data may help, but the remaining error may also reflect noise and a limited linear model; the curve cannot separate these explanations.

Learning-curve shading: one fold standard deviation, not a confidence interval. The fixed alpha was selected on these training data; this curve is descriptive.

5. Critical slices and error cases

BMI groups use training tertiles 24.2000 and 27.9000; intervals are right-closed. These are relative groups, not clinical categories.

bmi_slice	n	MAE
Low	38	34.490
Middle	23	47.433
High	28	50.288

The high BMI group has the largest test MAE (50.29, n=28). Group counts are small and unequal, so this is a descriptive difference on one split, not a population fairness or clinical finding.

row_ID	actual	predicted	abs_error	bmi	bmi_slice
56	52.000	205.195	153.195	30.200	High
77	200.000	74.619	125.381	23.000	Low
211	70.000	180.323	110.323	29.800	High

The three largest absolute errors range from 110.32 to 153.20 target units. Rows 56 and 211 have high BMI but low actual targets and are overpredicted; row 77 has low BMI but a high actual target and is underpredicted. People with similar measured inputs can still have different outcomes, and the model may miss unmeasured factors or nonlinear patterns. These are possible explanations, not verified causes. I do not change the model or alpha after inspecting these cases.

6. Discussion and limitations

Regression coefficients are conditional associations, not treatment effects. This is observational data without random assignment. Unmeasured confounding, correlated predictors, measurement error, and model assumptions can affect their signs and sizes. Regularization deliberately shrinks coefficients. Changing a feature would not necessarily cause the predicted outcome change.

This experiment uses 442 observations, one random split, and ten baseline features. Only 89 rows support the test diagnostics, and subgroup counts are smaller. There is no external validation, prediction interval, or formal heteroscedasticity test. CV is reused for selection, so the best CV score may be optimistic; its fold standard deviation is not a significance test. Simple linear models may miss nonlinear relationships and omitted variables. These results cannot support clinical decisions or causal claims.

7. Reproducibility and sources

Use Python 3.12 and requirements.txt. Install with python -m pip install -r requirements.txt, then run python -m nbconvert --execute --to notebook --inplace main.ipynb. The notebook recreates results.csv and this report. Checks verify the split, scaler training means, baseline mean, OLS agreement, all three test metrics and four figures. AI assistance is disclosed in AI_USE.md.

Data: scikit-learn.org/stable/datasets/toy_dataset.html#diabetes-dataset; original: www4.stat.ncsu.edu/~boos/var.select/diabetes.html. Influence: statsmodels.org/stable/generated/statsmodels.stats.outliers_influence.OLSInfluence.cooks_distance.html.